2 Algebra

What students need to learn:

Ref	Content descriptor	Concepts and skills
A a	Distinguish the different roles played by letter symbols in algebra, using the correct notation	• Use notation and symbols correctly
A b	Distinguish in meaning between the words 'equation', 'formula', **'identity'** and 'expression'	• Write an expression • Select an expression/**identity**/ equation/formulae from a list
A c	Manipulate algebraic expressions by collecting like terms, by multiplying a single term over a bracket, and by taking out common factors, **multiplying two linear expressions, factorise quadratic expressions including the difference of two squares and simplify rational expressions**	• Manipulate algebraic expressions by collecting like terms • Multiply a single term over a bracket • Use instances of index laws, including use of **fractional, zero and negative** powers, and powers raised to a power • Factorise algebraic expressions by taking out common factors • Write expressions to solve problems • Use algebraic manipulation to solve problems • **Expand the product of two linear expressions** • **Factorise quadratic expressions** • **Factorise quadratic expressions using the difference of two squares** • **Simplify rational expressions by cancelling, adding, subtracting, and multiplying**

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A d	Set up and solve simple equations **including simultaneous equations in two unknowns**	• Set up simple equations • Rearrange simple equations • Solve simple equations • Solve linear equations, with integer coefficients, in which the unknown appears on either side or on both sides of the equation • Solve linear equations that include brackets, those that have negative signs occurring anywhere in the equation, and those with a negative solution • Solve linear equations in one unknown, with integer or fractional coefficients • **Find the exact solutions of two simultaneous equations in two unknowns** • **Use elimination or substitution to solve simultaneous equations** • **Interpret a pair of simultaneous equations as a pair of straight lines and their solution as the point of intersection** • **Set up and solve a pair of simultaneous equations in two variables**
A e	**Solve quadratic equations**	• **Solve simple quadratic equations by using the quadratic formula** • **Solve simple quadratic equations by factorisation and completing the square**
A f	Derive a formula, substitute numbers into a formula and change the subject of a formula	• Derive a formula • Use formulae from mathematics and other subjects • Substitute numbers into a formula • Substitute positive and negative numbers into expressions such as $3x^2 + 4$ and $2x^3$ • Change the subject of a formula **including cases where the subject is on both sides of the original formula, or where a power of the subject appears**
A g	Solve linear inequalities in one or **two** variables, and represent the solution set on a number line **or coordinate grid**	• Solve simple linear inequalities in one variable, and represent the solution set on a number line • Use the correct notation to show inclusive and exclusive inequalities • **Show the solution set of several inequalities in two variables on a graph**

Higher

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A h	Use systematic trial and improvement to find approximate solutions of equations where there is no simple analytical method of solving them	• Use systematic trial and improvement to find approximate solutions of equations where there is no simple analytical method of solving them • Understand the connections between changes of sign and location of roots
A i	Generate terms of a sequence using term-to-term and position-to-term definitions of the sequence	• Recognise sequences of odd and even numbers • Generate simple sequences of numbers, squared integers and sequences derived from diagrams • Describe the term-to-term definition of a sequence in words • Find a specific term in a sequence using the position-to-term and term-to-term rules • Identify which terms cannot be in a sequence
A j	Use linear expressions to describe the n^{th} term of an arithmetic sequence	• Find the n^{th} term of an arithmetic sequence • Use the n^{th} term of an arithmetic sequence
A k	Use the conventions for coordinates in the plane and plot points in all four quadrants, including using geometric information	• Use axes and coordinates to specify points in all four quadrants in 2-D **and 3-D** • Identify points with given coordinates • Identify coordinates of given points (NB: Points may be in the first quadrant or all four quadrants) • Find the coordinates of points identified by geometrical information in 2-D **and 3-D** • Find the coordinates of the midpoint of a line segment • Calculate the length of a line segment

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A l	Recognise and plot equations that correspond to straight-line graphs in the coordinate plane, including finding gradients	• Draw, label and scale axes • Recognise that equations of the form $y = mx + c$ correspond to straight-line graphs in the coordinate plane • Plot and draw graphs of functions • Plot and draw graphs of straight lines with equations of the form $y = mx + c$ • Find the gradient of a straight line from a graph • **Find the gradient of lines given by equations of the form $y = mx + c$** • **Analyse problems and use gradients to interpret how one variable changes in relation to another**
A m	**Understand that the form $y = mx + c$ represents a straight line and that m is the gradient of the line and c is the value of the y- intercept**	• **Interpret and analyse a straight line graph** • **Understand that the form $y = mx + c$ represents a straight line** • **Find the gradient of a straight line from its equation**
A n	**Understand the gradients of parallel lines**	• **Explore the gradients of parallel lines and lines perpendicular to each other** • **Write down the equation of a line parallel or perpendicular to a given line** • **Select and use the fact that when $y = mx + c$ is the equation of a straight line then the gradient of a line parallel to it will have a gradient of m and a line perpendicular to this line will have a gradient of $-\frac{1}{m}$** • **Interpret and analyse a straight line graph and generate equations of lines parallel and perpendicular to the given line**

Higher

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A o	**Find the intersection points of the graphs of a linear and quadratic function, knowing that these are the approximate solutions of the corresponding simultaneous equations representing the linear and quadratic functions**	• **Solve exactly, by elimination of an unknown, two simultaneous equations in two unknowns, one of which is linear in each unknown, and the other is linear in one unknown and quadratic in the other, or where the second equation is of the form $x^2 + y^2 = r^2$** • **Find approximate solutions to simultaneous equations formed from one linear function and one quadratic function using a graphical approach** • **Select and apply algebraic and graphical techniques to solve simultaneous equations where one is linear and one quadratic**
A p	**Draw, sketch, recognise graphs of simple cubic functions, the**	• **Plot graphs of simple cubic functions, the reciprocal function $y = \frac{1}{x}$ with $x \neq 0$**